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Mismatch between quality and engagement in myasthenia gravis–related videos: a cross-sectional study on Chinese social media platforms

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Abstract

As short-video platforms increasingly play a prominent role in disseminating public health information in China, evaluating the quality of their health-related content is crucial. This study aims to systematically assess the quality, reliability, and content characteristics of videos about myasthenia gravis (MG) on China's mainstream short-video platforms (Douyin and Red Note). Using "myasthenia gravis" as the keyword, the top 100 videos returned by the default ranking on each platform were retrieved. After excluding duplicate and irrelevant videos, 174 videos, including 98 from Douyin and 76 from Red Note, were included in this cross-sectional content analysis. Transparency, overall informational quality, and general reliability were assessed using the Journal of the American Medical Association (JAMA) Benchmark criteria, Global Quality Score (GQS), and modified DISCERN instrument (mDISCERN), respectively. Video characteristics, uploader identities, content themes, and user engagement indicators were analyzed using nonparametric statistical methods. Physicians were the primary content source, accounting for 74.1% of the included videos. The most frequently addressed topics were treatment and management (47.1%) and clinical symptoms and presentation (40.2%), whereas diagnostic criteria and methods were addressed less frequently (8.0%). The median JAMA, GQS, and mDISCERN scores were 3.0 (interquartile range [IQR], 3.0–3.0), 3.0 (IQR, 3.0–4.0), and 2.0 (IQR, 2.0–3.0), respectively. Douyin videos had significantly higher raw numbers of likes, comments, saves, and shares than Red Note videos (all $P<0.001$). Douyin showed a more favorable distribution of JAMA scores, whereas the median GQS and mDISCERN scores were identical between platforms, despite statistically significant differences in their score distributions. Across uploader groups, significant differences were observed in JAMA and GQS scores, but not in mDISCERN scores ($P=0.091$). Hospitals and public health organizations had the highest median numbers of likes, saves, and shares, whereas videos posted by patients and their families had the highest median number of comments. Correlations between assessment scores and user engagement indicators were generally weak (Spearman's r ranging from −0.15 to 0.37). Content related to MG on Chinese short-video platforms generally exhibits moderate quality, with relatively favorable transparency but limited general reliability, and less frequent coverage of diagnostic criteria and examination methods. Content from professional medical sources tends to be better structured and more transparent, while patient-perspective content

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may generate greater comment engagement. Future efforts should strengthen collaboration between healthcare professionals and patients to improve the accessibility and comprehensibility of clinically reviewed MG-related content on short-video platforms.

Keywords Douyin · Global Quality Score · Short videos · Public health · Red Note

Abbreviations

MG	Myasthenia gravis
JAMA	Journal of the american medical association benchmark criteria
GQS	Global Quality Score
mDISCERN	modified DISCERN
IQR	Interquartile range
TCM	Traditional Chinese medicine
CNNIC	China internet network information center
FcRn	Neonatal Fc receptor

Introduction

Myasthenia gravis (MG) is a chronic autoimmune disorder of the neuromuscular junction, in which pathogenic mechanisms involve antibody-mediated impairment of neuromuscular transmission. Clinically, MG is characterized by fluctuating and fatigable skeletal muscle weakness[1]. The disease commonly presents with ocular symptoms such as ptosis and diplopia, but may also involve bulbar, limb, and respiratory muscles, leading to dysarthria, dysphagia, limb weakness, or respiratory compromise. Myasthenic crisis is generally defined as worsening MG that results in respiratory failure requiring ventilatory support, often accompanied by severe bulbar weakness, and represents one of the most serious complications in MG management[2]. Because early symptoms of MG may be subtle, fluctuating, and easily misattributed to other conditions or age-related changes, diagnostic delay and misdiagnosis remain common. In one previous study, diagnostic delays of 2 and 5 years were reported in 26% and 13% of patients with MG, respectively[3]. Delayed diagnosis may postpone the initiation of appropriate treatment and increase the risk of disease progression, crisis, and impaired quality of life. Therefore, clear, reliable, and accessible patient education on early recognition, timely referral, long-term treatment, and self-management is important for both clinical care and public health.

Although MG remains a relatively uncommon disease, accumulating epidemiological evidence has drawn increasing attention to its disease burden and evolving population profile. Previous studies have reported annual MG incidence rates of approximately 0.3–2.8 per 100,000 person-years and prevalence estimates of about 25 per 100,000 population[4,5]. A systematic review further showed substantial geographic heterogeneity in MG incidence and prevalence, suggesting that population structure, diagnostic capacity, and healthcare accessibility may influence observed epidemiological patterns[6]. In the context of population aging, late-onset MG has become increasingly relevant. Recent national registry data from Turkey also suggest a shift toward late-onset MG and indicate that the classic bimodal age-sex distribution may be attenuated in some populations, supporting the need to interpret MG epidemiology in the context of aging societies[7]. In China, a nationwide study estimated an age- and sex-standardized MG incidence of 0.68 per 100,000 person-years[8]. Cross-regional data have shown geographic differences in MG prevalence and treatment response[9], while mortality analyses have revealed regional disparities and an increasing mortality burden between 2013 and 2020[10]. Chinese cohort studies have also suggested that late-onset MG is relatively common[11,12]. Together, these findings indicate that MG should not be viewed solely as a specialist-managed rare disease, but also as a condition with important implications for long-term care, follow-up, regional healthcare equity, and patient education.

The information needs of patients with MG extend beyond pharmacological treatment. Patients and caregivers also require information on symptom recognition, diagnostic pathways, prognosis, daily living, treatment-related adverse effects, and the psychosocial burden of long-term disease management. International studies have increasingly examined MG from the perspectives of patient experience and digital health. The MyRealWorld MG study, an international digital observational study using patient-reported outcomes, highlights the value of real-world patient perspectives in evaluating treatment experience, symptom burden, quality of life, and psychosocial impact[13]. In addition, Legg et al. used cross-platform content analysis to identify unmet needs among patients with MG and caregivers, showing that online discussions involve treatment, symptoms, diagnosis, investigations, and daily living[14]. These studies suggest that digital platforms are not only channels for health-information seeking, but also spaces where patients' uncertainties and unmet informational needs become visible. For a disease such as MG, which is characterized by symptom fluctuation, diagnostic delay, and long-term management needs, it is important to examine whether online health information adequately covers early recognition, diagnostic evaluation, and evidence-based management.

Short-video platforms have become important entry points for health-information seeking in China. Compared with traditional text-based websites or long-form video platforms, short videos are visually intuitive, rapidly disseminated, narrative-driven, and shaped by algorithmic recommendation. These features may expand the reach of health education, but may also increase the visibility of emotionally appealing, experience-based, or highly interactive content. Previous studies across different diseases and platforms have reported incomplete information, insufficient source disclosure, and variable quality in online health videos[15–18]. In the field of MG, Meral et al. evaluated the educational value and content quality of MG-related videos on YouTube, providing important evidence on English-language MG video content[19]. However, Chinese short-video platforms differ from YouTube in language environment, user demographics, content-production patterns, recommendation mechanisms, and healthcare communication context. Douyin is one of the leading short-video platforms in China and is characterized by broad user reach and algorithm-driven dissemination. Red Note, also known as Xiaohongshu, combines social networking, experience sharing, and lifestyle-oriented content, where patients and caregivers may express illness experiences and healthcare-seeking narratives through note-style posts and videos. To date, little is known about how MG-related information is presented on major Chinese short-video platforms or how content quality relates to audience engagement.

Therefore, this cross-sectional study systematically assessed the content and presentation characteristics of MG-related short videos on Douyin and Red Note. Specifically, we evaluated the distribution of videos across platforms and uploader identities; the coverage of themes including symptoms, diagnostic evaluation, treatment and management, prognosis, and daily living; the transparency, overall quality, and general reliability reflected by JAMA, GQS, and mDISCERN scores; and the alignment between quality metrics and user engagement indicators. By characterizing these patterns, this study may provide a basis for future MG-specific content validation, clinician-patient co-created education, and platform-level strategies for improving the visibility, presentation, and accessibility of reliable MG-related educational content.

Methods

Study design

This was a cross-sectional content analysis of publicly available myasthenia gravis (MG)-related short videos on Douyin and Red Note. The study was designed to evaluate the presentation characteristics, thematic coverage, structural quality, general reliability, and user engagement of MG-related content on two major Chinese short-video platforms. The unit of analysis was an individual video[20–23].

Search strategy and data collection

Video searches were conducted on August 31, 2025, in Tianjin, China, using a Matebook D14 [with Intel(R) Core(TM) i5-10210U CPU @ 1.60 GHz (2.11 GHz)] and the Microsoft Edge browser (Version 140.0.3485.81). To minimize the influence of personalized recommendations, browser cache, cookies, and search history were cleared before the searches. A newly registered Red Note account linked to an unused phone number was used for the search on Red Note, whereas Douyin videos were accessed without logging into a personal account because public viewing was available.

The keyword "重症肌无力" (myasthenia gravis) was entered into each platform. The top 100 videos returned by the default platform ranking were collected from Douyin and Red Note, respectively. This strategy was used to approximate the videos that ordinary users would be most likely to encounter when actively searching for MG-related information within these platforms[15,24].

Videos were excluded if they were: (1) purely advertisements; (2) unrelated to MG; (3) not in Mandarin Chinese; or (4) duplicated on the same platform. A total of 200 videos were initially identified. Among them, 26 were excluded, including 2 purely advertising videos, 17 videos unrelated to MG, 3 videos not in Mandarin Chinese, and 4 duplicate videos. Finally, 174 videos were included in the analysis, comprising 98 Douyin videos and 76 Red Note videos. The video screening and selection process is shown in Fig. 1.

Data extraction

For each included video, the following information was extracted: platform, video link, video duration, uploader identity, communication format, background environment, visual information delivery, subtitle status, content themes, therapeutic interventions mentioned, and user engagement indicators. Video duration was recorded and analyzed in seconds.

User engagement indicators included the numbers of likes, comments, saves, and shares recorded on the date of data collection. These indicators were analyzed as raw platform interaction counts.

Uploader and content classification

Uploaders were classified according to their account identity and identifiable professional background. Medically affiliated accounts included hospitals and public health organizations, physicians, and medical students. Non-medically affiliated accounts included other public service organizations, science communicators, and patients or their families. Detailed classification criteria are provided in Supplementary Table 2.

Video presentation characteristics were classified according to communication format, background environment, visual information delivery, and subtitle status. Communication format was categorized as physician monologue, ward round, outpatient consultation, or other format. Background environment was categorized as hospital background, virtual background, or other background. Visual information delivery was categorized as animated graphics, static schematics, or text-only presentation. Subtitle status was recorded as subtitles present or absent. Detailed classification criteria are provided in Supplementary Table 3.

MG-related content themes were coded according to eight core topics: (1) etiology and pathophysiology; (2) disease examination methods; (3) diagnostic criteria and methods; (4) differential diagnosis; (5) treatment and management; (6) disease prognosis; (7) clinical symptoms and presentation; and (8) daily living status and quality of life. The category of clinical symptoms and presentation included MG-related symptoms explained by medical professionals as well as symptom descriptions based on patient or caregiver experience. Content themes were coded as non-mutually exclusive multilabel variables; therefore, a single video could involve multiple themes, such as clinical symptoms, diagnostic evaluation, and treatment and management.

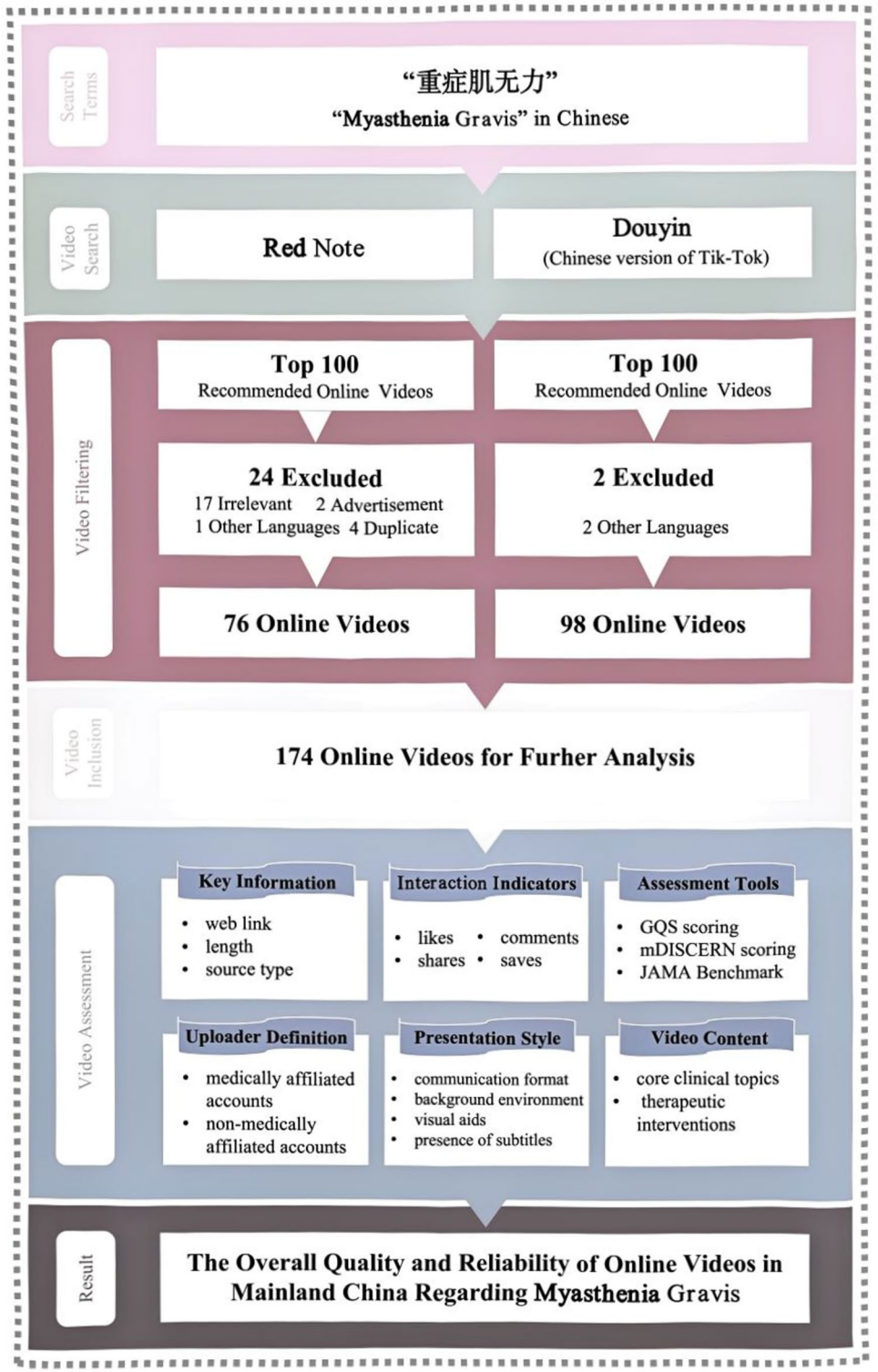


Fig. 1 Video search and screening process for myasthenia gravis.

Therapeutic interventions mentioned in the videos were also recorded, including acetylcholinesterase inhibitors, corticosteroids, immunosuppressants, biologic or targeted therapies, thymectomy, thymic radiotherapy, rehabilitation therapy, herbal decoctions, and acupuncture or moxibustion.

Quality and reliability assessment

Three commonly used instruments for evaluating online health information were applied to the included videos: the Journal of the American Medical Association (JAMA) Benchmark criteria, the Global Quality Score (GQS), and the modified DISCERN instrument (mDISCERN).

The JAMA Benchmark criteria were used to assess transparency and basic credibility of online health information across four domains: authorship, attribution, disclosure, and currency. Each item was scored as 1 for "yes" and 0 for "no," giving a total score ranging from 0 to 4, with higher scores indicating greater transparency.

The GQS was used to assess the overall informational quality, flow, and potential usefulness of the videos for patient education. The GQS is scored on a 1–5 scale, with higher scores indicating better overall informational quality and usefulness.

The mDISCERN instrument was used to assess the general reliability and balance of health information. It included five binary items, assessing whether the video was clear, concise, and understandable; whether valid sources were cited; whether the content was presented in a balanced and unbiased manner; whether additional sources of information were listed for patient reference; and whether areas of uncertainty were mentioned. Each item was scored as 1 for "yes" and 0 for "no", giving a total score ranging from 1 to 5, with higher scores indicating greater general reliability.

These instruments were used to assess transparency, overall informational quality, and general reliability. They were not designed to directly determine the clinical accuracy of individual MG-related statements, assess guideline concordance, identify MG-specific misinformation, or evaluate viewer knowledge acquisition, health-seeking behavior, or clinical outcomes. Detailed scoring criteria are provided in Supplementary Tables 4–7.

Evaluation process and inter-rater agreement

Two licensed physicians with clinical experience in MG independently evaluated all included videos. Before formal assessment, the two evaluators received standardized training on the JAMA, GQS, and mDISCERN scoring criteria and completed a calibration exercise using 10 sample videos to ensure consistent interpretation of the scoring standards[25–28].

During formal evaluation, each evaluator independently recorded the JAMA, GQS, and mDISCERN scores for each video. When discrepancies occurred, the two evaluators first discussed the source of disagreement. If consensus could not be reached, a third senior evaluator reviewed the video and made the final decision according to the predefined scoring criteria. Inter-rater agreement before consensus was assessed using Cohen's kappa coefficient.

Statistical analysis

Categorical variables were summarized as numbers and percentages. Because video duration and user engagement indicators were skewed and the score variables were ordinal or non-normally distributed, continuous variables were summarized as medians and interquartile ranges.

Comparisons of continuous variables between two independent groups were performed using the Mann-Whitney U test. For mutually exclusive grouping variables, such as platform and uploader identity, comparisons of continuous variables across multiple groups were performed using the Kruskal-Wallis test, with adjustment for multiple comparisons when appropriate. For non-mutually exclusive content-theme variables, descriptive summaries were provided, and comparisons across content categories were interpreted cautiously. Categorical

variables were compared using the chi-square test or Fisher's exact test. Spearman correlation analysis was used to assess associations between quality assessment scores and user engagement indicators. All statistical tests were two-sided, and P values < 0.05 were considered statistically significant.

All statistical analyses were performed using IBM SPSS Statistics version 27.0.1.

Ethical considerations

This study analyzed only publicly available videos from Douyin and Red Note. No non-public user information was accessed, and the researchers did not interact with video uploaders or viewers. All data were analyzed in aggregate, and no personally identifiable information was collected. According to institutional policy, formal ethics approval and individual informed consent were not required.

Results

Video characteristics

As shown in Table 1, a total of 174 videos were analyzed. The median (IQR) values for video duration, likes, comments, saves, and shares were 84.0 (57.0–131.5), 92.5 (21.3–319.8), 12.5 (2.0–48.8), 29.5 (8.0–130.8), and 21.5 (7.0–106.8), respectively. In terms of uploader identity, physicians contributed the largest proportion of videos (74.1%, 129/174), followed by patients and their families (12.6%, 22/174), science communicators (4.6%, 8/174), other public service organizations (4.0%, 7/174), hospitals and public health organizations (3.4%, 6/174), and medical students (1.1%, 2/174).

Because content themes were coded as non-mutually exclusive variables, a single video could be assigned to more than one content category. Treatment and management was the most frequently addressed topic (47.1%, 82/174), followed by clinical symptoms and presentation (40.2%, 70/174), patients' daily living status and quality of life (21.8%, 38/174), etiology and pathophysiology (16.7%, 29/174), differential diagnosis (16.7%, 29/174), disease prognosis (15.5%, 27/174), disease examination methods (10.3%, 18/174), while diagnostic criteria and methods were addressed less frequently (8.0%, 14/174). The median scores for JAMA, GQS, and mDISCERN were 3.0 (3.0–3.0), 3.0 (3.0–4.0), and 2.0 (2.0–3.0), respectively.

Furthermore, we compared video characteristics between the two platforms. Inter-rater agreement between the two evaluators was excellent, with kappa coefficients of 0.889 for GQS, 0.843 for mDISCERN, and 0.968 for JAMA. As shown in Table 2, video duration did not differ significantly between Red Note and Douyin. Douyin videos had significantly higher raw engagement counts than Red Note videos in terms of likes, comments, saves, and shares (all $P < 0.001$). For assessment scores, JAMA, GQS, and mDISCERN differed significantly between platforms ($P < 0.001$, $P = 0.025$, and $P = 0.012$, respectively). Douyin videos showed a higher distribution of JAMA scores than Red Note videos, although both platforms had a median JAMA score of 3.0. The two platforms also had the same median GQS and mDISCERN scores, despite statistically significant differences in their overall score distributions (Fig. 2).

To gain deeper insights, we analyzed video characteristics according to uploader identity (Table 3). Hospitals and public health organizations received the highest median numbers of likes, saves, and shares (181.0, 90.5, and 82.0, respectively), while videos uploaded by patients and their families received the highest median number of comments (91.5). Medical students produced the longest videos (median duration: 929.0 s), although this subgroup included only two videos. Further comparisons of assessment scores showed significant differences in JAMA and GQS scores across uploader groups, but not in mDISCERN scores. Physicians and hospitals and public health organizations had the highest JAMA scores (3.0 [3.0–3.0]). Regarding GQS scores, hospitals and public health organizations achieved the highest ratings (4.0 [3.3–4.0]). In contrast, mDISCERN scores did not

Table 1 Video characteristics.

Characteristic		*N* = 174
Uploader identity [*n*(%)]	Physicians	129(74.1)
	Hospitals and public health organizations	6(3.4)
	Medical students	2(1.1)
	Other public service organizations	7(4.0)
	Science communicators	8(4.6)
	Patients and their families	22(12.6)
Video content[n(%)]	Etiology and pathophysiology	29(16.7)
	Clinical symptoms and presentation	70(40.2)
	Diagnostic criteria and methods	14(8.0)
	Differential diagnosis	29(16.7)
	Treatment and management	82(47.1)
	Disease prognosis	27(15.5)
	Disease examination methods	18(10.3)
	Patients' daily living status and quality of life	38(21.8)
Therapeutic interventions[n(%)]	Acetylcholinesterase inhibitors	30(17.2)
	Corticosteroids	24(13.8)
	Immunosuppressants	34(19.5)
	Biologic/targeted therapies	4(2.3)
	Thymectomy	29(16.7)
	Thymic radiotherapy	7(4.0)
	Rehabilitation therapy	8(4.6)
	Herbal decoctions	24(13.8)
	Acupuncture and moxibustion	6(3.4)
Communication format[n(%)]	Physician monologue	104(59.8)
	Ward round	11(6.3)
	Outpatient consultation	21(12.7)
	Other format	38(21.8)
Background environment[n(%)]	Hospital	128(73.6)
	Virtual background	7(4.0)
	Other background	39(22.4)
Visual information delivery[n(%)]	Animated graphics	154(88.5)
	Static schematics	8(4.6)
	Text-only	12(6.9)
Subtitles status[n(%)]	Subtitled	160(92.0)
	Non-subtitled	14(8.0)
Duration, s		84.0(57.0–131.5)
Likes		92.5(21.3–319.8)
Comments		12.5(2.0–48.8)
Saves		29.5(8.0–130.8)
Shares		21.5(7.0–106.8)
JAMA score		3.0(3.0–3.0)
GQS score		3.0(3.0–4.0)
mDISCERN score		2.0(2.0–3.0)

Values are presented as n (%) or median (IQR), as appropriate. Duration is reported in seconds.

differ significantly among uploader groups ($P = 0.091$). Figure 3 provides detailed comparisons of assessment scores across different uploader groups.

Assessment of video content

We further analyzed video characteristics according to content categories. Most videos addressed treatment and management (47.1%, 82/174), followed by clinical symptoms and presentation (40.2%, 70/174), patients' daily living status and quality of life (21.8%, 38/174), etiology and pathophysiology (16.7%, 29/174), differential

Table 2 Video characteristics according to uploader identity.

Video features, median (IQR)	Red Note ($n=76$)	Douyin ($n=98$)	P
Duration, s	85.0(56.8–157.5)	84.0(57.0–125.0)	0.780
Likes	20.0(11.0–41.5)	245.0(96.0–546.0)	<0.001
Comments	2.0(0.8–11.0)	32.5(9.0–115.0)	<0.001
Saves	11.0(5.8–26.0)	85.0(14.8–228.8)	<0.001
Shares	10.5(5.0–24.8)	62.5(12.2–184.0)	<0.001
JAMA score	3.0(2.0–3.0)	3.0(3.0–3.0)	<0.001
GQS score	3.0(3.0–4.0)	3.0(3.0–4.0)	0.025
mDISCERN score	2.0(2.0–3.0)	2.0(2.0–2.0)	0.012

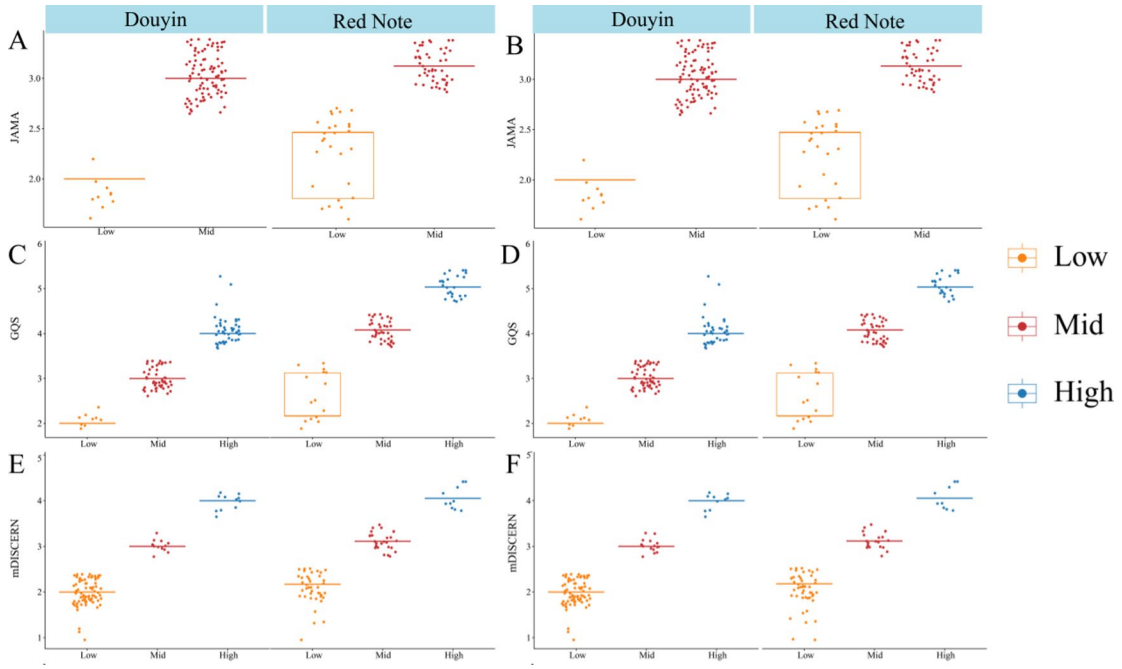


Fig. 2 Score distributions across platforms by two evaluators. Low: rating≤2; Mid: 2<rating<4; High: rating≥4. Panels A/C/E show the scores assigned by evaluator 1, and panels B/D/F show the scores assigned by evaluator 2. JAMA scores are shown in panels A and B, GQS scores in panels C and D, and mDISCERN scores in panels E and F.

diagnosis (16.7%, 29/174), disease prognosis (15.5%, 27/174), disease examination methods (10.3%, 18/174), and diagnostic criteria and methods (8.0%, 14/174). Basic data for different video content categories are presented in Table 4. Overall comparisons showed significant differences in JAMA, mDISCERN, and GQS scores across content categories ($P=0.049$, $P<0.001$, and $P<0.001$, respectively). Post-hoc comparisons showed that JAMA scores differed significantly between treatment and management and daily living status and quality of life ($P<0.01$). For mDISCERN, diagnostic criteria and methods scored significantly higher than daily living status and quality of life ($P<0.001$), and treatment and management also scored significantly higher than daily living status and quality of life ($P=0.014$) (Fig. 4).

Correlation analysis

We employed Spearman correlation analysis to examine relationships among various video variables. As shown in Figure 5, JAMA scores demonstrated correlation coefficients of 0.08 with mDISCERN and 0.43 with GQS, while showing correlations of -0.15, 0.10, -0.05, 0.15, and 0.18 with video duration, likes, comments, saves, and shares, respectively. The mDISCERN score correlated with GQS at 0.28, and with video duration, likes,

Table 3 Video characteristics by content category.

Video characteristic	Uploader group/identity						*P*
	Physicians (*n* = 129)	Hospitals and public health organizations (*n* = 6)	Medical students (*n* = 2)	Other public service organizations (*n* = 7)	Science communicators (*n* = 8)	Patients and their families (*n* = 22)	
Duration, s	81.0 (60.0–112.0)	154.5 (101.8–230.5)	929.0 (588.5–1269.5)	50.0 (10.0–74.5)	64.0 (55.3–151.3)	118.0 (48.5–214.3)	0.008
Likes	98.0 (25.0–365.0)	181.0 (50.5–520.0)	28.0 (24.5–31.5)	11.0 (5.0–19.0)	9.0 (3.5–21.5)	124.0 (22.3–288.5)	0.006
Comments	12.0 (3.0–37.0)	9.0 (3.0–44.3)	1.0 (1.0–1.0)	0.0 (0.0–6.0)	0.0 (0.0–6.0)	91.5 (15.8–187.8)	< 0.001
Saves	32.0 (10.0–149.0)	90.5 (38.3–246.3)	10.5 (9.8–11.3)	7.0 (3.5–10.0)	2.0 (1.0–8.3)	25.0 (4.8–71.3)	0.005
Shares	25.0 (8.0–117.0)	82.0 (46.5–274.3)	3.5 (2.8–4.3)	7.0 (2.5–11.5)	4.0 (1.0–8.8)	22.0 (7.0–80.0)	0.005
JAMA score	3.0 (3.0–3.0)	3.0 (3.0–3.0)	2.0 (1.5–2.5)	1.0 (1.0–2.0)	2.0 (2.0–2.8)	2.0 (2.0–2.8)	< 0.001
GQS score	3.0 (3.0–4.0)	4.0 (3.3–4.0)	3.0 (3.0–3.0)	2.0 (1.0–2.5)	3.0 (2.0–4.0)	3.0 (2.0–4.0)	0.008
mDISCERN score	2.0 (2.0–3.0)	2.0 (2.0–2.8)	2.5 (2.3–2.8)	3.0 (3.0–3.5)	2.0 (2.0–2.0)	2.0 (2.0–2.0)	0.091

Values are presented as median (IQR). Duration is reported in seconds.

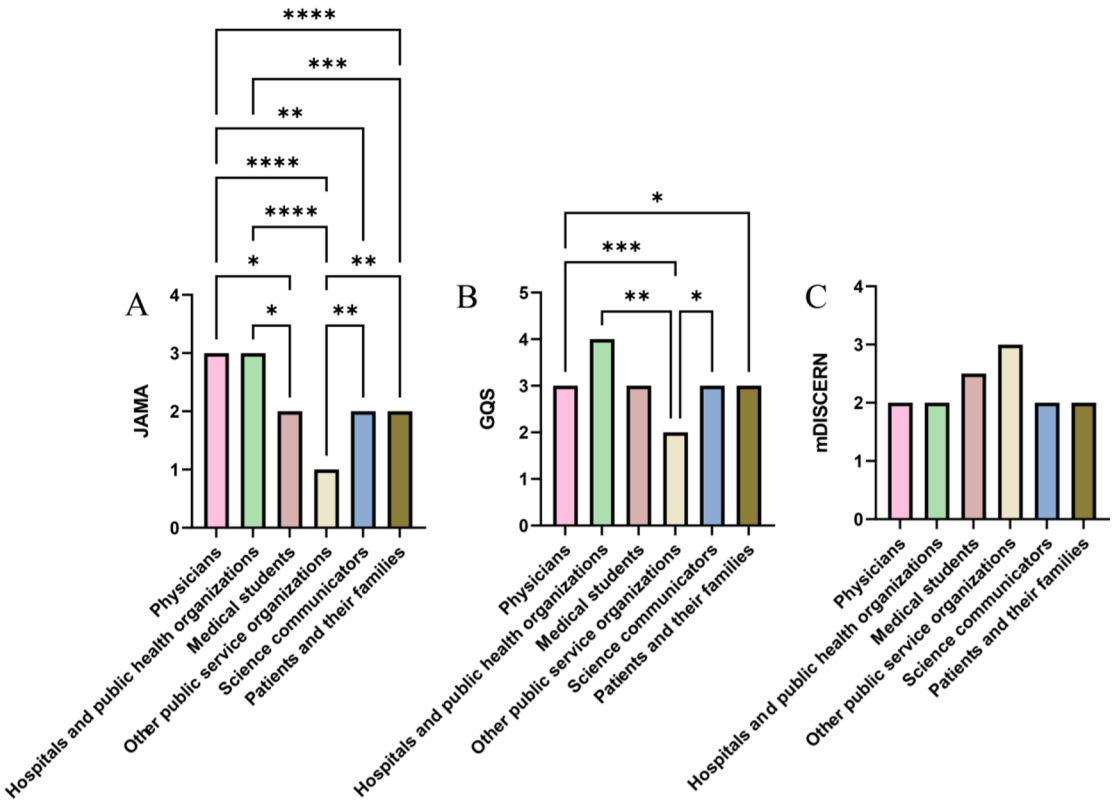


Fig. 3 Assessment scores across uploader groups. (**A**) JAMA scores. (**B**) GQS scores. (**C**) mDISCERN scores. **P* < 0.05; ***P* < 0.01; ****P* < 0.001; *****P* < 0.0001.

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Table 4 Video characteristics across platforms.

P	Etiology and pathophysiology (*n* = 29)	Disease prognosis (*n* = 27)	Clinical symptoms and presentation (*n* = 70)	Disease examination methods (*n* = 18)	Differential diagnosis (*n* = 29)	Patients' daily living status and quality of life (*n* = 38)	Treatment and management (*n* = 82)	Diagnostic criteria and methods (*n* = 14)	
0.581	103.0(76.0–179.0)	81.0(68.5–129.5)	88.0(56.3–142.3)	99.0(73.3–198.5)	84.0(65.0–128.0)	103.0(50.0–170.0)	92.0(62.0–152.5)	104.0(75.3–230.8)	Duration, s
0.760	93.0(18.0–561.0)	197.0(22.0–325.5	97.0(22.8–332.5)	40.0(17.3–169.0)	96.0(21.0–524.0)	153.5(30.3–307.3)	79.0(24.3–363.3)	68.0(22.8–279.8)	Likes
0.044	13.0(4.0–32.0)	30.0(6.0–94.0)	12.0(2.0–35.8)	3.5(1.0–11.3)	7.0(3.0–52.0)	44.0(6.8–181.5)	9.0(2.0–35.0)	10.5(1.3–32.3)	Comments
0.788	11.0(6.0–157.0)	53.0(8.5–177.5)	28.5(7.3–150.3)	16.5(7.3–45.5)	33.0(6.0–233.0)	40.0(7.8–109.0)	31.0(9.3–148.8)	18.5(9.8–187.8)	Saves
0.648	13.0(3.0–211.0)	46.0(8.5–244.5)	23.5(7.0–123.0)	7.5(4.0–22.5)	21.0(7.0–158.0)	27.5(7.3–105.5)	26.0(8.0–113.8)	21.0(5.5–331.8)	Shares
0.049	3.0(3.0–3.0)	3.0(3.0-3.0)	3.0(3.0-3.0)	3.0(3.0-3.0)	3.0(3.0-3.0)	3.0(2.0–3.0)	2.0(2.0–3.0)	3.0(3.0-3.0)	JAMA score
< 0.001	3.0(3.0–4.0)	4.0(3.0–4.0)	3.0(3.0–4.0)	3.0(3.0–3.8)	3.0(3.0–4.0)	3.0(3.0–4.0)	3.0(3.0-3.0)	3.5(3.0–4.0)	GQS score
< 0.001	3.0(2.0–3.0)	2.0(2.0–2.5)	2.0(2.0–3.0)	2.0(2.0–3.0)	2.0(2.0–3.0)	2.0(2.0–2.0)	3.0(3.0–4.0)	3.0(2.3–4.0)	mDISCERN score

Values are presented as median (IQR). Duration is reported in seconds. Content categories were non-mutually exclusive.

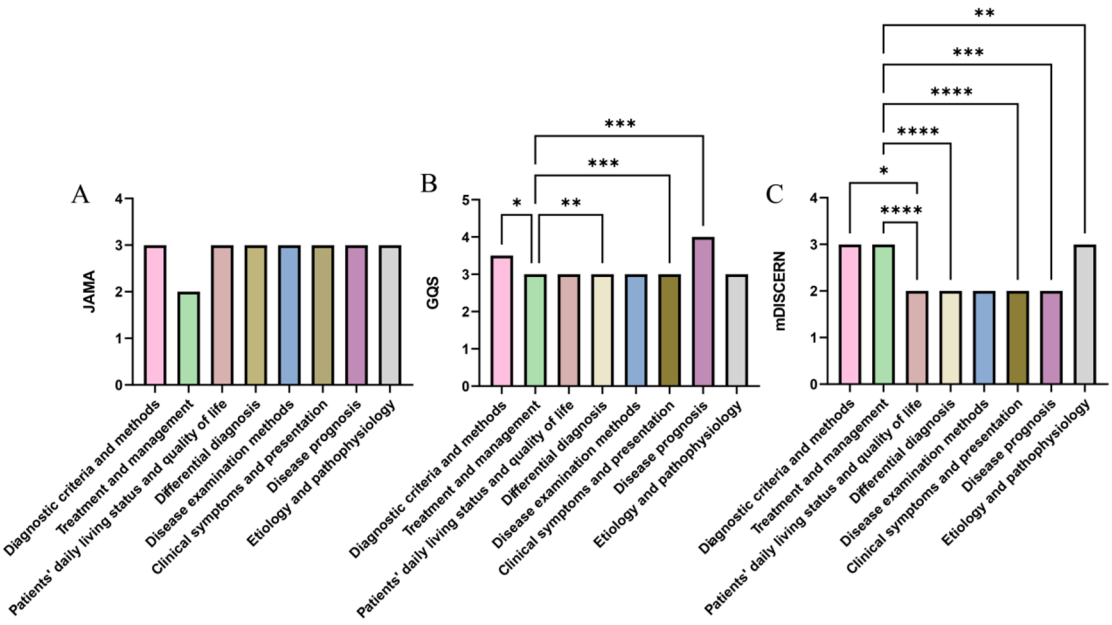


Fig. 4 Comparison of assessment scores across content categories. (**A**) JAMA scores. (**B**) GQS scores. (**C**) mDISCERN scores. *$P<0.05$; **$P<0.01$; ***$P<0.001$; ****$P<0.0001$.

comments, saves, and shares at 0.11, 0.13, 0.03, 0.19, and 0.17, respectively. GQS scores correlated with video duration, likes, comments, saves, and shares at 0.18, 0.31, 0.18, 0.37, and 0.35, respectively. Video duration showed correlation coefficients of 0.52, 0.20, 0.27, and 0.14 with likes, comments, saves, and shares, respectively. Likes demonstrated correlations of 0.53, 0.76, and 0.70 with comments, saves, and shares, respectively. Comments correlated with saves and shares at 0.63 and 0.58, respectively, while saves and shares showed a correlation coefficient of 0.92.

Discussion

MG, a complex autoimmune disorder, presents growing diagnostic and therapeutic challenges in an aging population[29]—a concern particularly relevant in China, which is undergoing rapid demographic aging. Within China's distinctive healthcare landscape, MG management is further complicated by an uneven distribution of specialized medical resources and regional disparities in diagnostic capacity[8]. Older adults often experience symptoms that overlap with age-related physiological decline, which can be easily misattributed to normal aging, thereby delaying diagnosis[9]. Meanwhile, the ongoing digital transformation in health communication has opened new pathways for patient education and disease management. Short-video platforms are increasingly serving as a vital bridge between professional expertise and public health literacy, especially in regions with limited access to conventional health communication channels. According to the latest data from the China Internet Network Information Center (CNNIC), as of June 2025, China had approximately 161 million Internet users aged 60 and above, representing an Internet penetration rate of about 52% in this age group[30]. Older adults are also increasingly turning to digital devices to seek health-related information[31]. This study provides a systematic analysis of MG-related short-video content on Chinese platforms. In response to the opportunities and heightened need for reliable health content, this research aims to characterize how MG is represented online and to establish an evidence-based foundation for enhancing digital health education. Overall, MG-related videos showed moderate informational quality and relatively favorable transparency, but limited general reliability, with comparatively infrequent coverage of diagnostic criteria and examination methods. Notably, quality assessment scores and user

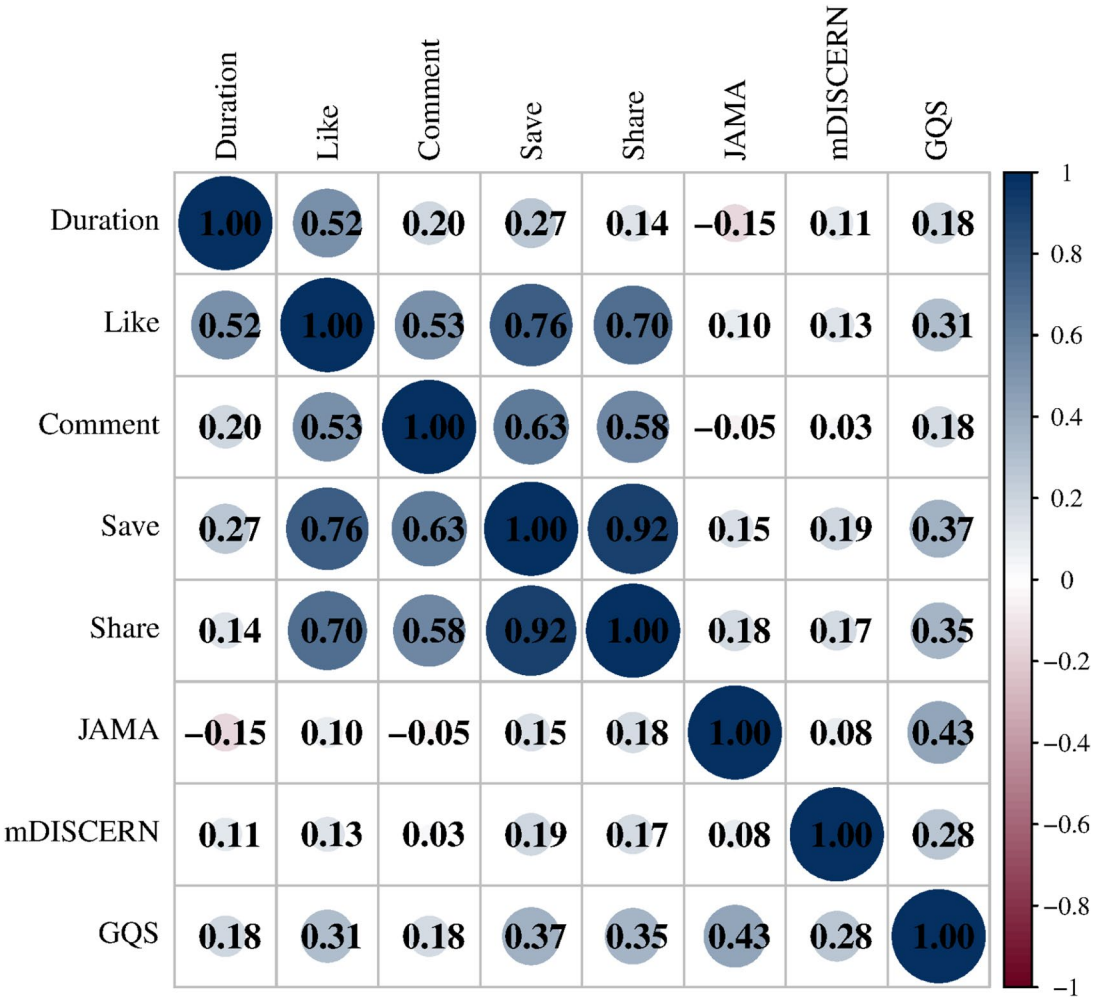


Fig. 5 Spearman correlation matrix among video characteristics, user engagement indicators, and assessment scores.

engagement indicators were not consistently aligned, suggesting a potential mismatch between information dissemination and reliability. This pattern is broadly consistent with previous MG-related digital-content studies reporting uneven coverage of diagnosis, treatment, symptoms, and daily-living concerns[14,19].

Overall video evaluation

Our analysis showed that MG-related short videos on Chinese platforms demonstrated generally acceptable informational quality but with important weaknesses in reliability. Across the 174 included videos, the median GQS, JAMA, and mDISCERN scores were 3.0 (IQR, 3.0–4.0), 3.0 (IQR, 3.0–3.0), and 2.0 (IQR, 2.0–3.0), respectively, indicating moderate overall informational quality, relatively favorable transparency, and limited general reliability. This pattern of limited general reliability is consistent with prior evaluations of medical content on open video platforms, where clinically themed videos often appear authoritative yet omit key elements such as evidence citation, adverse-effect warnings, and criteria for urgent medical review[27]. Importantly, the medically affiliated group accounted for 78.7% (137/174) of our sample, and physicians were the most prominent content source within this group (94.2%, 129/137). However, professional affiliation was not associated with significant differences in mDISCERN scores. This suggests that clinical expertise does not automatically translate into comprehensive patient education in the short video format. These findings support the need for structured

communication standards for MG-related short videos—not only to improve clarity and perceived quality, but also to ensure that safety-critical information is delivered in a balanced and reproducible way. From a public health perspective, insufficiently reliable information may increase the risk of misunderstanding disease management and delay appropriate care-seeking behaviors, particularly in populations with limited access to specialized neurological services.

The Spearman correlation analysis demonstrated that the three quality evaluation tools—GQS, mDISCERN, and JAMA—showed weak to moderate intercorrelations ($r = 0.08$–0.43), indicating that they capture distinct dimensions of video quality and reliability rather than overlapping constructs. Among audience-interaction indicators, the strongest associations were observed between likes, comments, saves, and shares ($r = 0.53$–0.92), reflecting coherent user behavior patterns in engagement. However, correlations between assessment scores and user engagement indicators ranged from −0.15 to 0.37. For example, mDISCERN and JAMA scores—which emphasize general reliability and transparency, respectively—showed little to no correlation with engagement variables, whereas GQS exhibited only modest positive associations (likes $r = 0.31$, saves $r = 0.37$, and shares $r = 0.35$). These findings indicate that content popularity should not be regarded as a proxy for better informational quality or general reliability.

This decoupling between information quality and audience engagement may stem from the inherently specialized nature of MG. As a rare autoimmune neuromuscular disease, its clinical terminology, symptom heterogeneity, and complex management strategies pose barriers to lay understanding. High-quality educational videos often prioritize completeness and accuracy, but this technical density can reduce accessibility for general viewers, given the generally limited health literacy among the public[32,33]—a phenomenon observed in other conditions with similar diagnostic complexity, such as thyroid eye disease[34]. Moreover, MG's fluctuating clinical course and chronic fatigue symptoms may drive patient and caregiver interest toward experiential or coping-related content, leading to engagement preferences not aligned with formal quality metrics. Overall, these findings highlight the need to improve medical communication design rather than solely pursuing content accuracy. Similar discrepancies between information quality and audience engagement have been reported in studies of health-related content on platforms such as YouTube and TikTok, suggesting that this challenge may be common across different digital health ecosystems[35,36].

Our analysis revealed significant heterogeneity in video quality and engagement outcomes depending on uploader identity. Among the six subgroups analyzed, significant differences were observed in JAMA and GQS scores ($P < 0.001$ and $P = 0.008$, respectively), but not in mDISCERN scores ($P = 0.091$). Physicians and hospitals/public health organizations had the highest median JAMA scores (3.0 [3.0–3.0]), whereas hospitals/public health organizations had the highest median GQS score (4.0 [3.3–4.0]). These findings are consistent with previous evaluations on TikTok, Kwai, and Bilibili, in which institutional or physician-produced content achieved higher transparency or overall quality scores[25,37,38]. Overall differences across uploader groups were also significant for likes, comments, saves, and shares (all $P < 0.01$). Hospitals and public health organizations had the highest median numbers of likes, saves, and shares, whereas videos posted by patients and their families had the highest median number of comments. The particularly high comment engagement observed for patient- and family-posted videos may reflect the conversational value of lived experience and experiential narratives. However, greater comment engagement should not be interpreted as evidence of higher informational quality, clinical accuracy, or educational effectiveness.

Video content

Thematic analysis revealed that treatment and management dominated MG-related video content (47.1%), followed by clinical symptoms and presentation (40.2%) and patients' daily living status and quality of life (21.8%). In contrast, diagnostic criteria and methods (8.0%) and disease examination methods (10.3%) were among the least frequently addressed topics. This pattern suggests a creator emphasis on symptom control and therapeutic options—areas of immediate concern to patients and caregivers—at the expense of early recognition and

diagnostic guidance. While this focus aligns with the demand for actionable knowledge, the limited coverage of diagnostic criteria and examination methods may constrain the usefulness of short videos as an entry point for early recognition and appropriate referral. Such delays are well-documented: in one international survey 30.9% of patients waited three years or more for accurate diagnosis[39]. Another retrospective analysis also indicated that, due to the diverse manifestations of MG, it is often misdiagnosed as other diseases[40]. Enhancing messaging around early diagnosis—particularly through clear descriptions of hallmark fluctuating symptoms such as fatigable ptosis, dysarthria, and dysphagia[41,42]—could raise disease awareness and facilitate timely referral to neurology specialists. From a public health perspective, improving awareness of early diagnostic features through accessible digital content may support earlier symptom recognition and timely referral to neurological services.

Given the prominence of treatment and management content in the MG-related short-video landscape, we further analyzed the specific therapeutic approaches represented. Our findings show that immunosuppressants (19.5%), acetylcholinesterase inhibitors (17.2%), and thymectomy (16.7%) were the most frequently discussed interventions. This distribution reflects a balanced emphasis on both pharmacological and surgical strategies, which aligns with established clinical practices in China[43]. In these practices, pharmacotherapy forms the cornerstone of long-term management, primarily through the use of acetylcholinesterase inhibitors and corticosteroids, while thymectomy is considered a key disease-modifying procedure—particularly in patients with acetylcholine receptor antibody-positive generalized MG[44]. The notable emphasis on thymectomy in the videos may also be influenced by the substantial proportion of content created by hospitals and physicians, who often highlight standard interventional treatments available in tertiary care settings. In contrast, novel biologic and targeted therapies—including complement inhibitors (eculizumab, ravulizumab) and neonatal Fc receptor (FcRn) antagonists (e.g., efgartigimod)—were scarcely represented (2.3%). This gap likely reflects their limited availability, significant cost, and relatively recent introduction in China[45–47]. These findings indicate that recent advances in MG immunotherapy were less frequently represented in the sampled videos.

Interestingly, our analysis revealed a notable presence of traditional Chinese medicine (TCM) in MG-related short videos, with herbal decoctions appearing in 13.8% and acupuncture or moxibustion in 3.4% of all included videos. This reflects the ongoing integration of TCM perspectives into digital patient education—a trend consistent with national policies advocating the incorporation of digital technologies into the preservation and innovation of TCM[48]. The potential of these TCM modalities is supported by growing research[49]. Herbal formulas are noted for their multi-component and multi-target nature, with reported mechanisms relevant to MG including modulation of the gut microbiota and regulation of immune-inflammatory responses[50–52]. Additionally, the anti-inflammatory role of acupuncture in autoimmune conditions is well-established[53]; a recent meta-analysis further found that adjunctive acupuncture was associated with improved treatment response and reductions in MG-related clinical scores, although the quality of the available evidence remains limited[54]. These findings provide a scientific context for the presence of TCM-related content on Chinese short-video platforms; however, the present study did not assess the clinical accuracy, evidentiary basis, or therapeutic value of individual TCM-related statements in the included videos.

Limitations and future directions

Several limitations should be considered. First, this was a cross-sectional content analysis conducted at a single time point. Because short-video platforms are highly dynamic, video ranking, visibility, and engagement may change with platform policies, recommendation algorithms, and trending topics. The findings therefore reflect MG-related videos available on the search date rather than long-term trends.

Second, only Douyin and Red Note were included. The findings may not be generalizable to other Chinese or international platforms, such as Bilibili, WeChat Channels, Weibo, and YouTube. Although browser cache, cookies, and search history were cleared and a newly registered Red Note account was used, search rankings may still have been influenced by geographic location, account status, platform algorithms, and real-time popularity. The retrieved videos may therefore differ from those encountered by individual users.

Third, user engagement indicators in this study, including likes, comments, saves, and shares, were analyzed as raw platform interaction counts. We were unable to obtain information on video views, account verification status, recommendation exposure, or platform-specific interaction mechanisms. Therefore, engagement metrics could not be fully standardized and should not be used to compare reach-adjusted dissemination performance across platforms. Higher engagement may reflect user interest, emotional resonance, experiential value, or platform recommendation mechanisms, rather than higher informational quality, better educational effectiveness, or greater clinical value.

Fourth, the JAMA Benchmark criteria, GQS, and mDISCERN assess information transparency, overall quality, and general reliability, but do not directly evaluate the clinical accuracy, guideline concordance, MG-specific completeness, or misinformation contained in individual statements. Accordingly, higher scores cannot be equated with clinical accuracy, and lower scores do not necessarily indicate incorrect or harmful information. Future studies should incorporate specialist review and MG-specific, guideline-based assessment criteria.

Fifth, viewer-level outcomes were not evaluated. Platform interactions do not indicate whether viewers improved their knowledge, recognized warning signs, changed care-seeking behavior, or adhered more closely to treatment. Surveys, experimental viewing studies, and pre-post assessments are needed to examine the educational and behavioral effects of MG-related short videos.

Sixth, subgroup findings should be interpreted cautiously because some uploader categories contained few videos and group sizes were imbalanced. In addition, content themes were non-mutually exclusive; therefore, theme-related results represent descriptive comparisons of overlapping content features rather than comparisons among independent groups.

Future research could further develop MG-specific evaluation frameworks for online health information. Such frameworks should cover early symptom recognition, diagnostic pathways, examination methods, differential diagnosis, crisis warning signs, long-term treatment, and self-management. In addition, expert review by neurologists, neuroimmunologists, or neuromuscular specialists should be incorporated to complement the limitations of JAMA, GQS, and mDISCERN in assessing clinical accuracy and guideline concordance. Future MG-related health communication could also explore clinician-patient co-created short videos that combine professional medical knowledge, patient experience, and accessible communication. At the platform level, future efforts could explore verified labels, topic collections, or expert-reviewed content recommendation mechanisms to improve the visibility, comprehensibility, and accessibility of reliable MG-related educational content. Longitudinal studies are also needed to examine how content quality, thematic coverage, and user engagement change over time.

Conclusion

This cross-sectional content analysis evaluated the information presentation, thematic coverage, structural quality, general reliability, and user engagement of MG-related short videos on Douyin and Red Note. MG-related content on Chinese short-video platforms generally showed moderate overall informational quality and relatively favorable transparency, but limited general reliability. Treatment and management and clinical symptoms and presentation were the most frequently addressed topics, whereas diagnostic criteria and methods and disease examination methods were less frequently represented. Quality assessment scores were not consistently aligned with user engagement, indicating that higher engagement should not be regarded as a proxy for better informational quality. Because JAMA, GQS, and mDISCERN do not directly assess clinical accuracy, guideline concordance, or educational impact, these findings should be interpreted as a characterization of information presentation, thematic coverage, structural quality, general reliability, and user engagement rather than as evidence of clinical correctness or viewer benefit. Future studies should incorporate MG-specific expert validation and explore clinician-patient co-created educational content to improve the visibility, accessibility, and comprehensibility of clinically reviewed MG-related information on short-video platforms.

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Data availability All data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval and consent to participate This study did not involve human participants or animal subjects. All data were obtained from publicly accessible online platforms (Douyin and Red Note) and were analyzed in an aggregated manner without interaction with individuals or access to private identifiable information. According to institutional policy, formal ethical approval and informed consent were not required.

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